

FactorEngine Reproduction Report - Draft V2

A sharper interpretation of the live Kimi/Moonshot 300-iteration run on real CSI300/CSI500 A-share data. This is a separate draft artifact and does not replace the current report.

Executive Takeaway

VALIDATION LIFT

8.8x

seed fitness to best evolved fitness

LIVE KIMI ACCEPTED

250/300

83.3% of evolution iterations

BEST FRONTIER REACHED

iter 99

201 later iterations did not improve best reward

CSI300 MODEL IC DELTA

-0.0042

augmented minus baseline on 2017-2024 test

CSI500 MODEL IC DELTA

+0.0009

small IC lift, but backtest quality worsened

CSI500 SINGLE-FACTOR IC

+0.0167

standalone FE-evolved signal

What is solid

The live macro-agent path is now real, not simulated: **250** accepted Kimi/Moonshot mutations were applied through the same path used by evolution. The engine lifted validation fitness from **+0.072** to **+0.636**, a **8.8x** improvement, with a best factor built around `llm`, `turnover`, `volscale`.

What the current report should say more carefully

The augmented model is not a clean out-of-sample win. On CSI300 it lowers model IC from **+0.0158** to **+0.0117**. On CSI500 it slightly raises model IC from **+0.0094** to **+0.0103**, but annual return and Sharpe deteriorate versus baseline. The right conclusion is not "Kimi failed"; it is "validation-only elite selection overfit, and the portfolio layer did not convert standalone signal into robust PnL."

Run Diagnostics

Item	Value	Interpretation
Panel evolved	csi300	Evolution ran on CSI300 and transferred to CSI500.
Iterations	300	Requested live macro-agent run.
Accepted live LLM mutations	250	Connectivity worked; fallback handled transient/unparseable calls.
Provider / model	kimi-cn / kimi-k2.6	Moonshot China path.
Endpoint	https://api.moonshot.cn/v1	Direct HTTPS route used by the evolution call path.
Temperature	1.0	Model-specific Moonshot constraint.
Runtime	14.6h (52,492s)	175.0s per iteration.
Evaluations / nodes	3602 / 554	Micro-search and tree size.
Best transforms	llm , turnover , volscale	Current live run: LLM idea plus turnover and volatility scaling.
Declared tuned params	w1=0.752 , w2=0.3795 , w3=0.0552 , vol_window=27	Only these entered Bayesian search.

The best validation frontier was reached by iteration **99**. The remaining **201** iterations explored variants without improving the global best reward, so future runs should add early stopping or switch objective once the frontier plateaus.

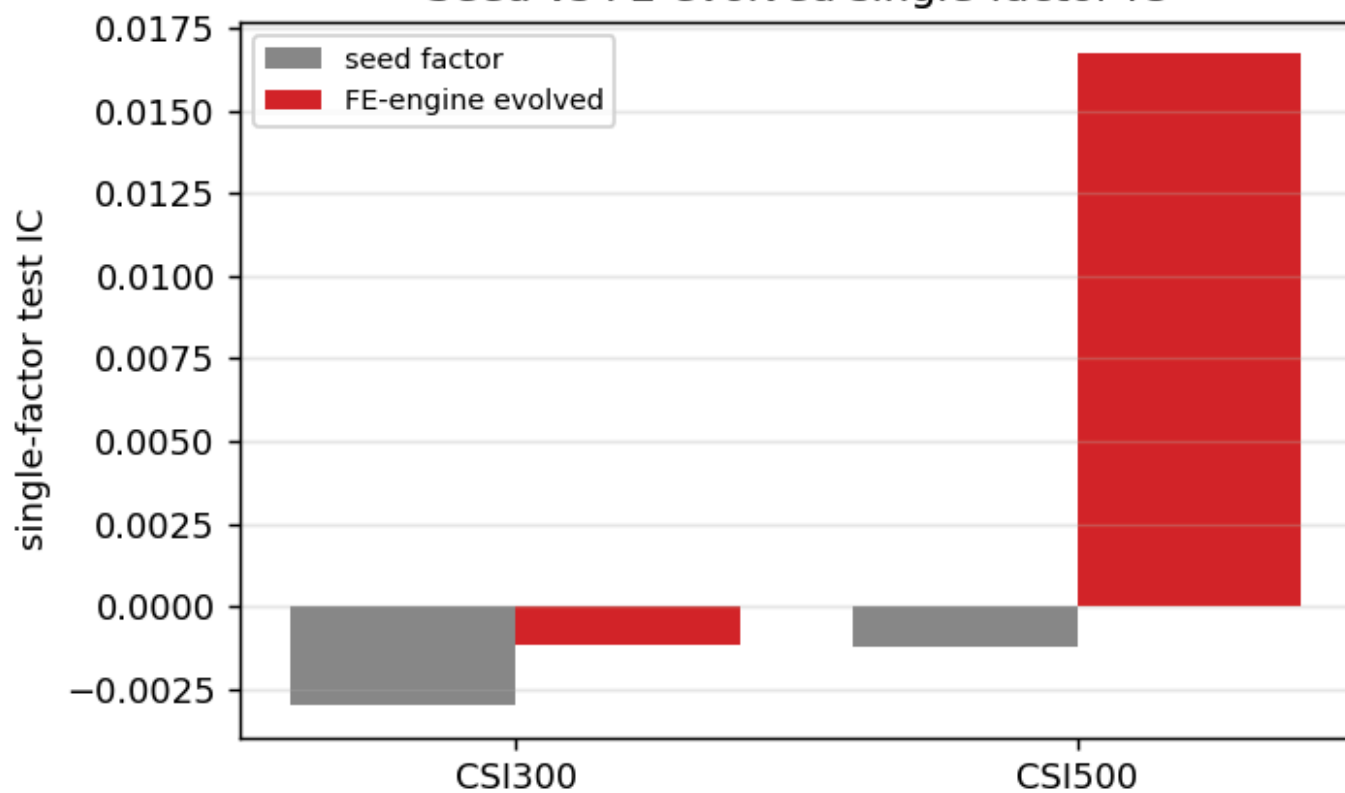
Record iteration	Best reward
1	+0.1017
2	+0.4033
11	+0.4050
20	+0.5060
57	+0.5143
58	+0.5162
62	+0.5182
67	+0.5205
70	+0.5273
76	+0.5761
99	+0.5765

Result 1 - Single-Factor Mining

The evolution loop does produce a better factor, especially on CSI500. This matters because it separates the factor-mining result from the later model-combination result.

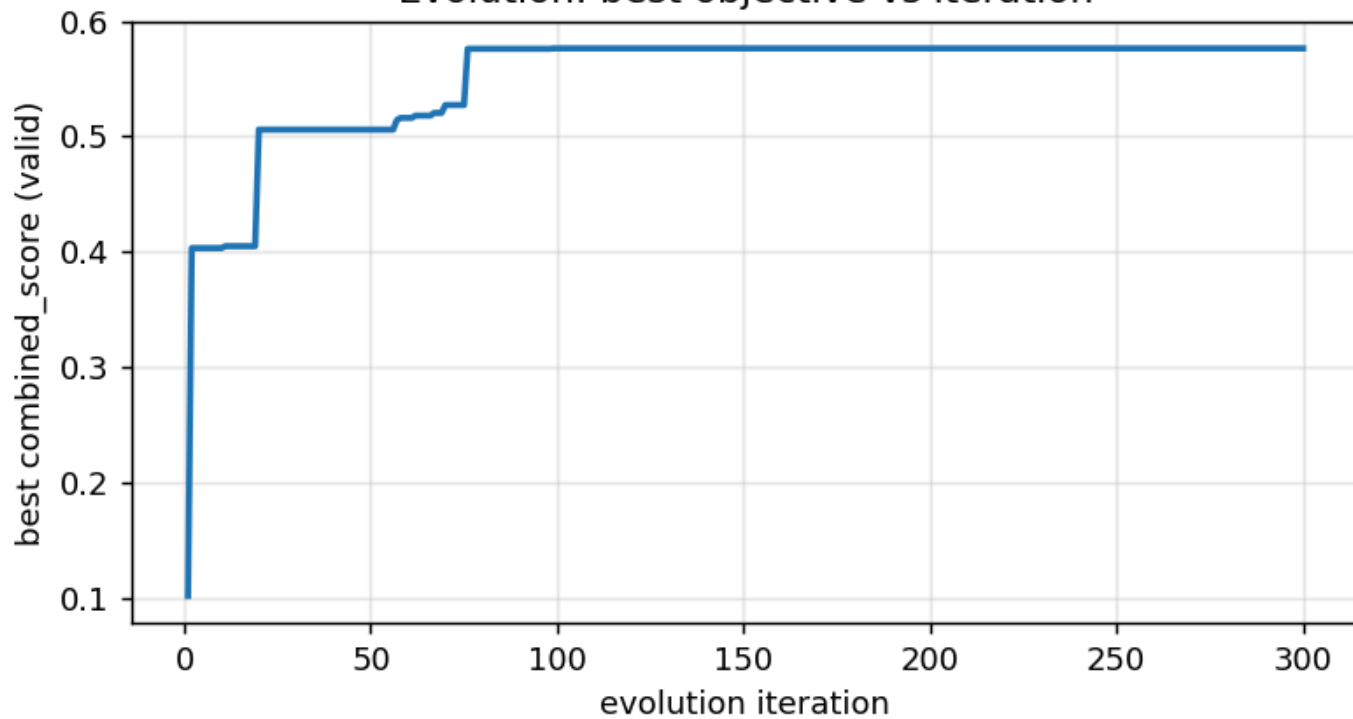
Universe / factor	IC	ICIR	RankIC	RankICIR	Fitness
CSI300 - Seed	-0.0030	-0.031	-0.0020	-0.016	-0.024
CSI300 - Paper artifact	-0.0033	-0.029	-0.0035	-0.029	-0.031
CSI300 - FE-evolved	-0.0012	-0.010	+0.0051	+0.040	+0.018
CSI500 - Seed	-0.0012	-0.014	-0.0049	-0.047	-0.031
CSI500 - Paper artifact	-0.0099	-0.094	-0.0164	-0.148	-0.126
CSI500 - FE-evolved best standalone	+0.0167	+0.138	+0.0387	+0.303	+0.249

Seed vs FE-evolved single-factor IC



Single-factor IC comparison

Evolution: best objective vs iteration



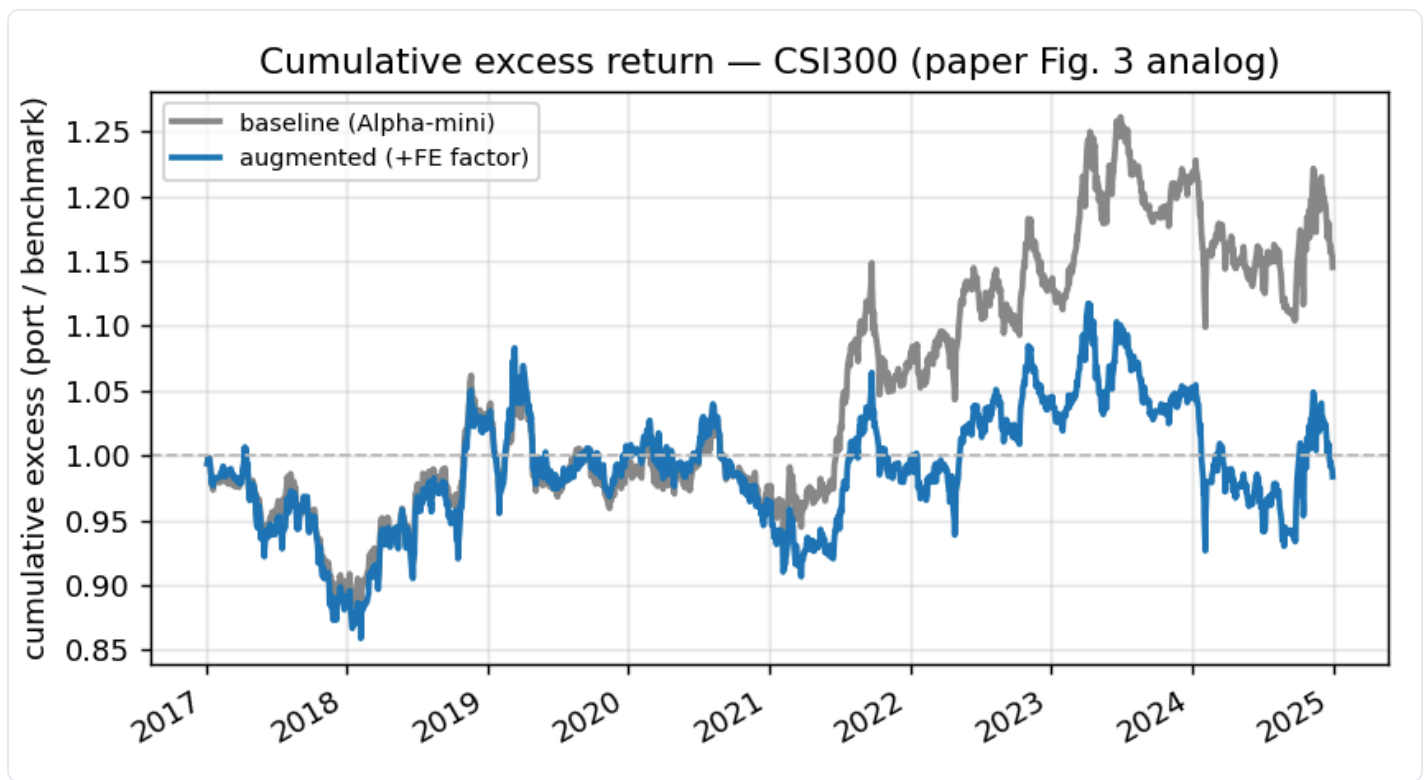
Validation convergence

Result 2 - Multi-Factor Integration Is The Weak Link

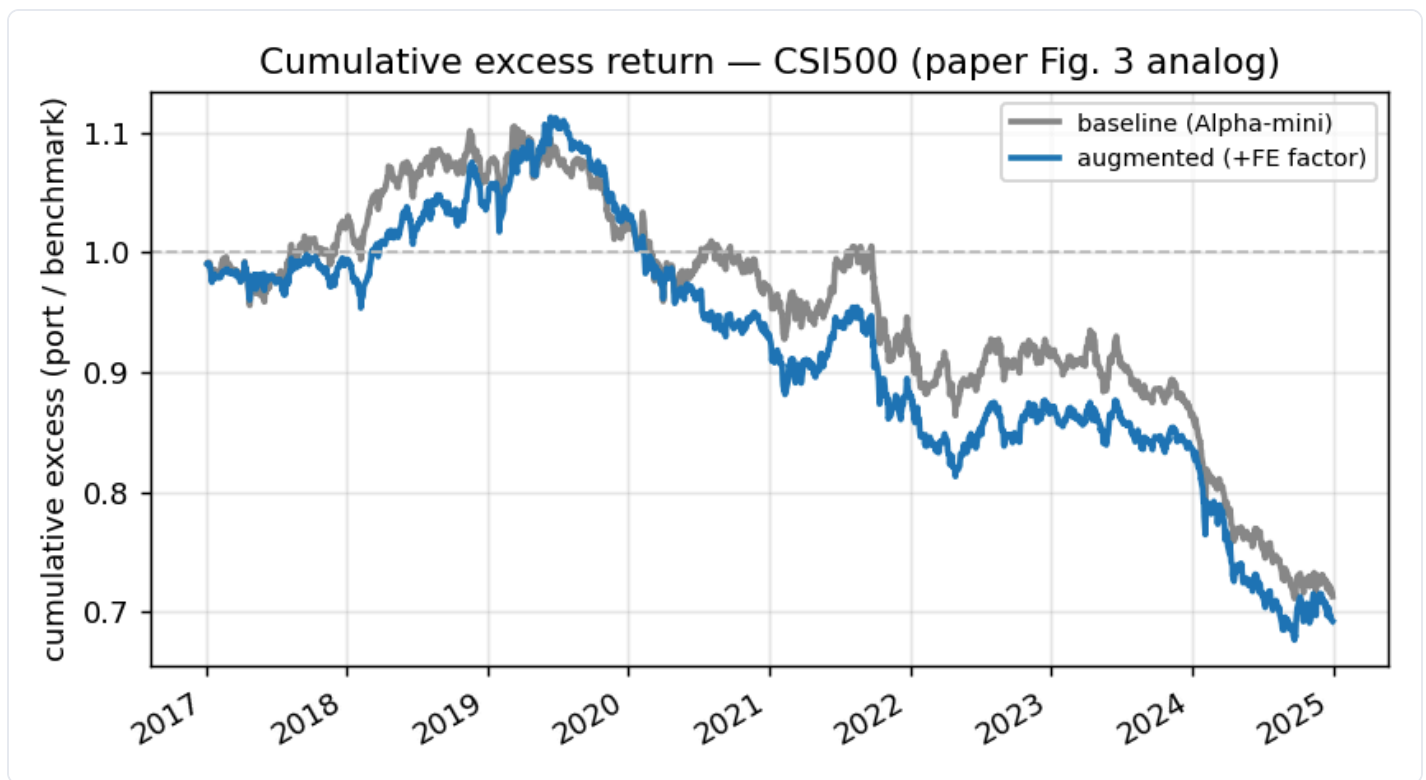
The augmented model merges elite FE factors with Alpha158-128 features. The corrected reading is mixed: CSI300 gets worse; CSI500 model IC improves slightly, but the backtest does not improve versus the baseline. This points to integration and selection, not raw LLM connectivity, as the next bottleneck.

Universe / model	IC	RankIC	AR	SR	MDD	Features	Top features
CSI300 - Alpha158-128	+0.0158	+0.0185	+3.96%	+0.28	+37.46%	128	STD30:46, KLOW:44, ROC5:38, ROC60:38, VWAP0:31, KLEN:30
CSI300 - GPLearn arm IC winner	+0.0172	+0.0182	+2.43%	+0.22	+40.24%	136	-
CSI300 - + FE elites	+0.0117	+0.0150	+1.92%	+0.20	+45.48%	133	KLOW:32, STD30:24, MAX5:18, fe_elite_2:18, KLEN:14, HIGH0:13
CSI500 - Alpha158-128	+0.0094	+0.0173	-5.52%	-0.10	+58.04%	128	ROC60:9, KLOW:7, STD30:7, VWAP0:6, STD10:6, KSFT:5
CSI500 - GPLearn arm	+0.0099	+0.0145	-6.80%	-0.15	+59.55%	136	-
CSI500 - + FE elites IC winner, not portfolio winner	+0.0103	+0.0238	-5.88%	-0.11	+57.65%	133	KLOW:5, VWAP0:4, ROC5:3, MA5:3, MAX5:3, RSV5:3

Universe	Delta IC	Delta RankIC	Delta AR	Delta SR	Delta annual cost
CSI300	-0.0042	-0.0035	-2.03%	-0.086	+0.07%
CSI500	+0.0009	+0.0065	-0.36%	-0.019	+0.09%



CSI300 equity curve: baseline vs augmented

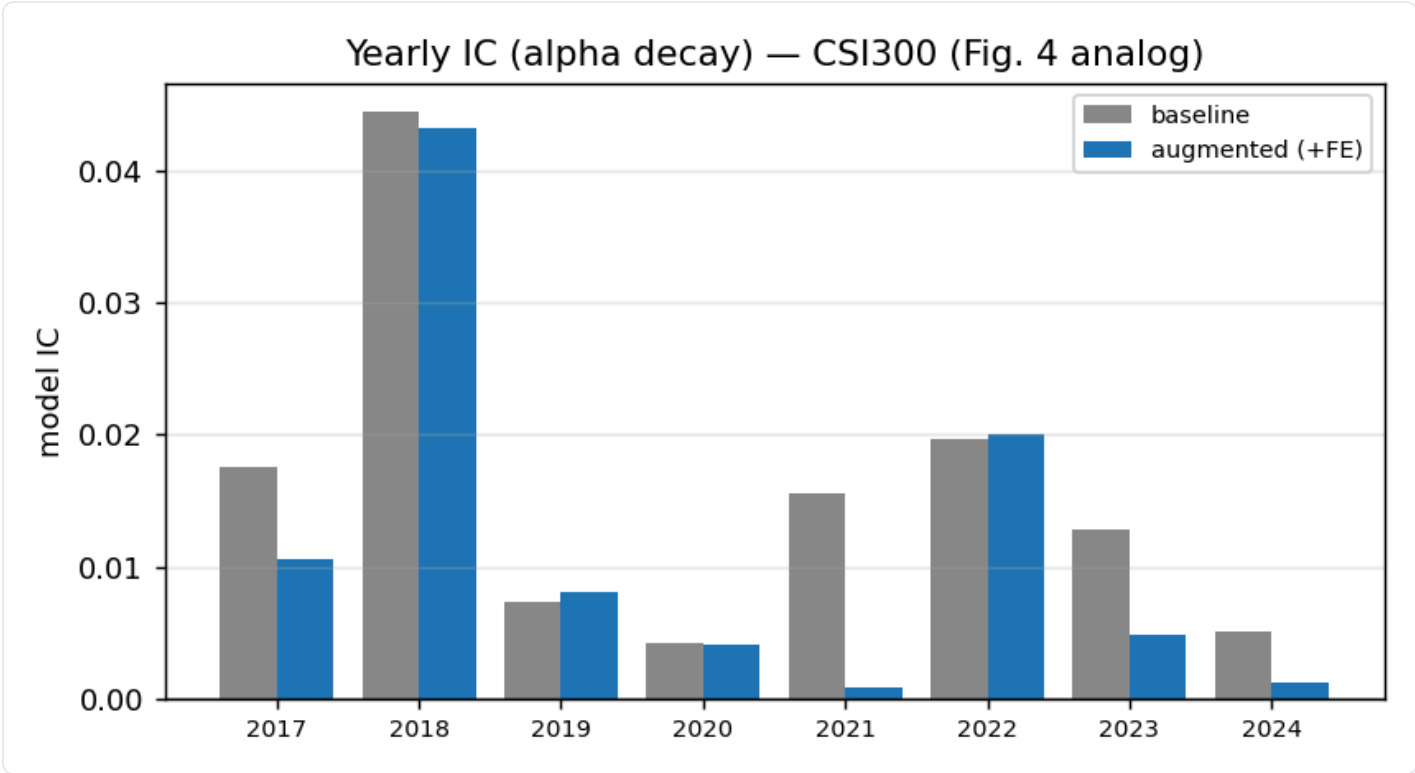


CSI500 equity curve: baseline vs augmented

Where The Overfit Shows Up

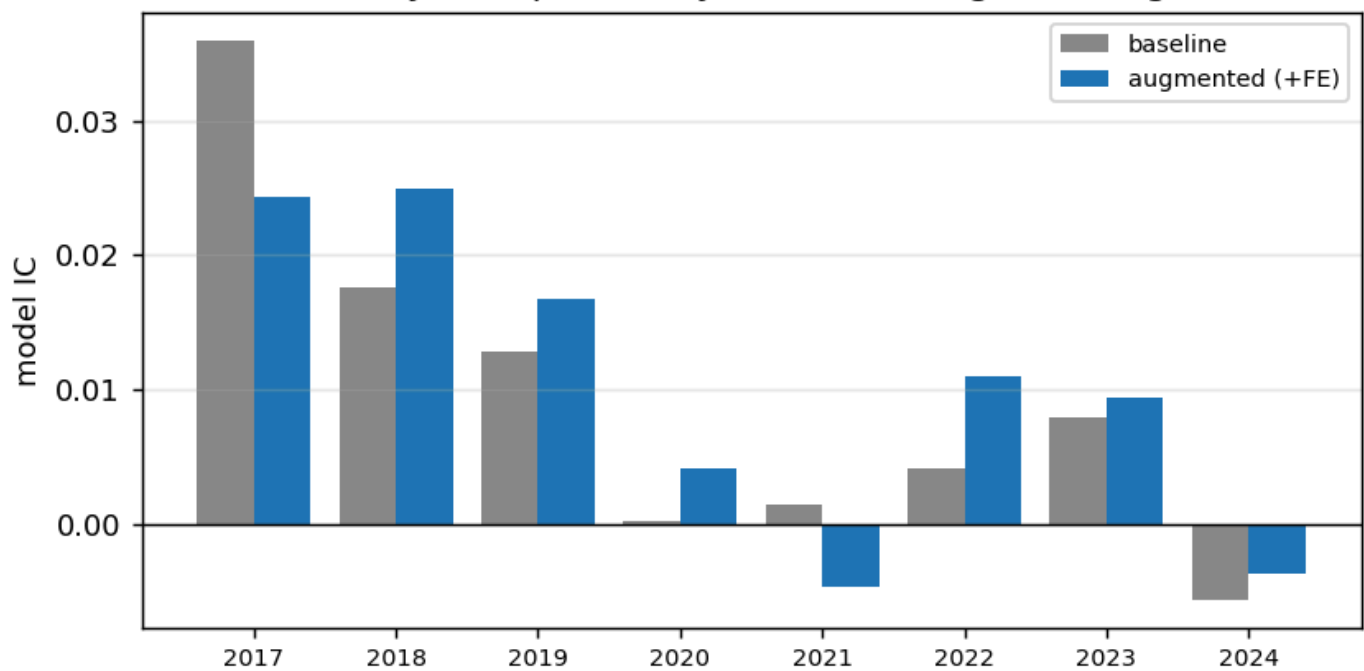
Validation lift is large, but the test behavior is uneven by year and by universe. The CSI500 standalone factor has signal, while the elite bundle and model weighting are fragile.

Universe	Augmented IC better	Augmented IC worse	Worst year	Worst IC delta
CSI300	2/8 years better	6/8 years worse	2021	-0.0146
CSI500	6/8 years better	2/8 years worse	2017	-0.0117



CSI300 yearly IC

Yearly IC (alpha decay) — CSI500 (Fig. 4 analog)



CSI500 yearly IC

Parameter Contract Warning

Kimi introduced knobs outside the Bayesian search contract

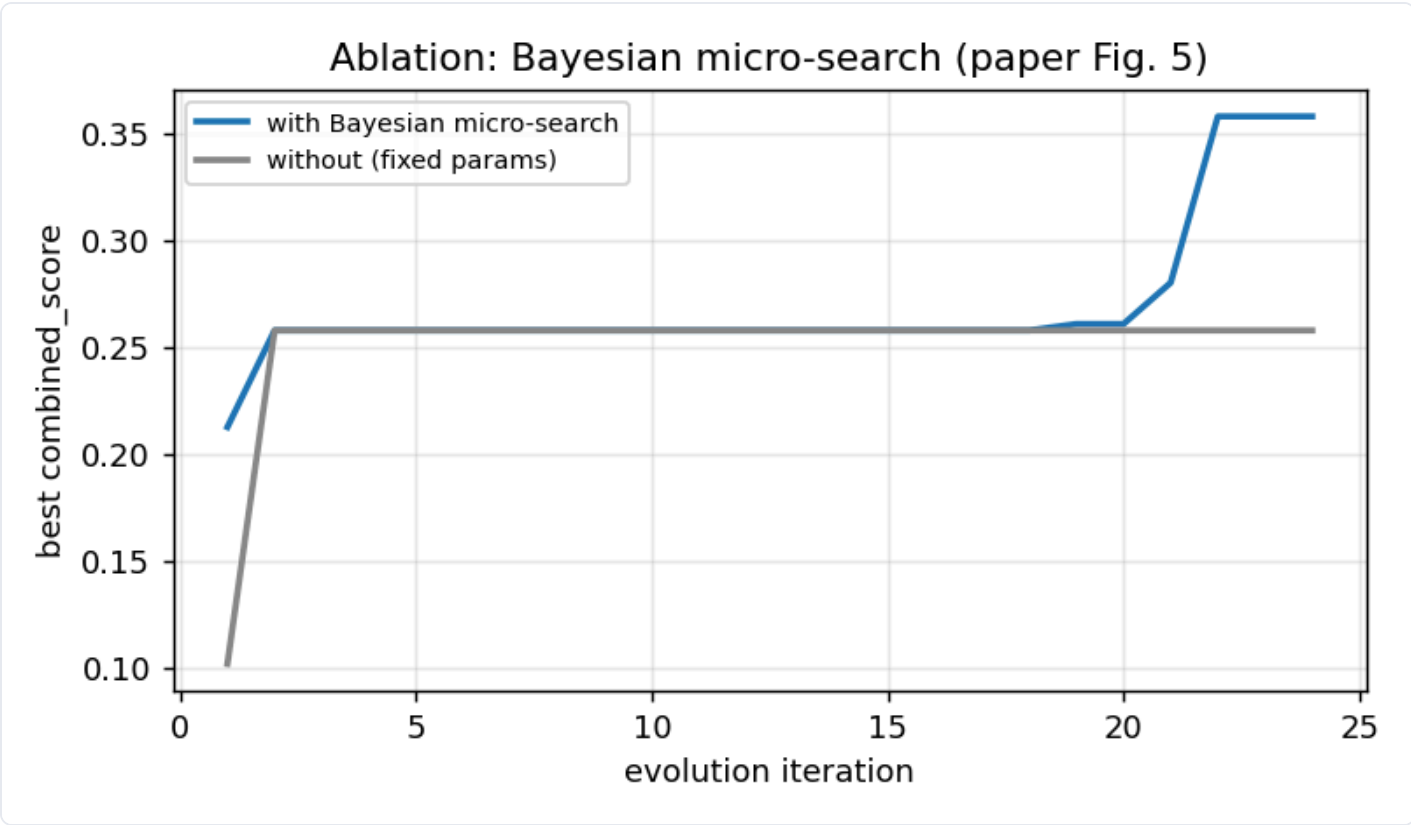
The best code reads these parameters: `epsilon`, `smooth_window`, `turnover_decay`, `vol_window`, `w1`, `w2`, `w3`, `w4`, `w_turnover`. The saved Bayesian-tuned params are: `w1=0.752`, `w2=0.3795`, `w3=0.0552`, `vol_window=27`. Undeclared runtime defaults are: `smooth_window`, `turnover_decay`, `w4`, `w_turnover`. Extra declared-but-unused params are: None.

This is a constructive bug/insight rather than a fatal flaw. The macro-agent learned useful structure, but some choices stayed as hard-coded defaults. For the next run, every `parameters.get(...)` key should be auto-scanned and either forced into `###Parameters` or rejected before evaluation.

Ablation Reading

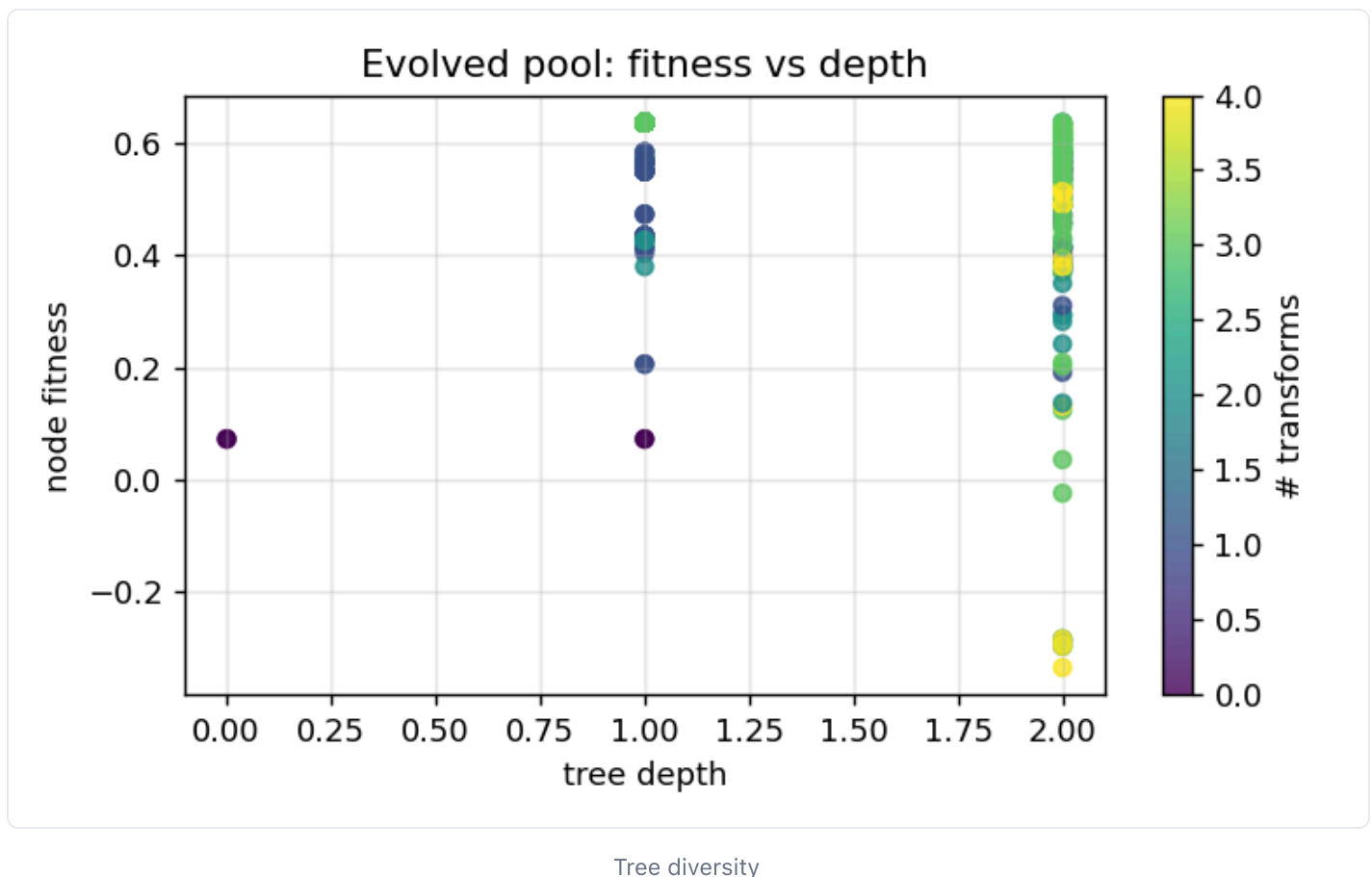
Bayesian search still appears load-bearing, but the island result warns that more diversity can increase validation fitness while harming transfer if selection remains validation-only.

Control	Best reward	Best fitness	Best transforms
Bayesian ON	+0.358	+0.445	ewm, ranknorm, turnover
Bayesian OFF	+0.258	+0.308	proposal_volscaled_momentum



Bayesian micro-search ablation

Islands	Mining fitness	CSI300 test RankIC	CSI300 test fitness	Transforms
1 island(s)	+0.318	+0.0022	+0.068	proposal_amihud_reversal, proposal_gk_lowvol, proposal_overnight_intraday, proposal_pv_divergence, proposal_volscaled_momentum
2 island(s)	+0.374	-0.0124	-0.103	midprice, volscale



Recommended V3 Protocol

- **Change elite selection.** Select factors by multi-window, cross-universe robustness: validation fitness, test-like rolling subwindows, sign consistency, and degradation penalties.
- **Penalize complexity.** Add a penalty for parameter count, transform count, and undeclared defaults. A factor with hidden knobs should lose score until the knobs are tuned.
- **Cap redundancy.** Reject or downweight factors highly correlated with Alpha158 features or existing elite factors. The model needs orthogonal alpha, not more versions of the same turnover/liquidity exposure.
- **Make objective portfolio-aware.** Include turnover, cost, and yearly stability in the micro objective, not only IC/ICIR on one validation regime.
- **Use plateau-aware compute.** This run found the frontier by iteration 99. After 30-50 stagnant iterations, switch to a new seed, new split, or robustness objective instead of spending another 200 iterations on the same frontier.
- **Tighten the LLM prompt and parser.** Require one mutation theme per response, all tunables declared in JSON ranges, no lookahead, and a short rationale tied to CSI300/CSI500 microstructure.

Draft Conclusion

The strongest defensible conclusion is: **the live Kimi/Moonshot macro-agent can drive FactorEngine evolution and lift validation fitness, but the current validation-only elite selection overfits.** The factor itself is not useless; CSI500 standalone OOS IC is **+0.0167**, and CSI500 augmented model IC is slightly positive versus baseline. The failure is that the selected elite bundle does not translate reliably into out-of-sample portfolio quality. The next improvement should target the selection/objective layer before increasing iteration count or model size.